

Engineering Mathematics III (ENGE702)

C Hassell Sweatman

Lecture 22: The Wave Equation Derivation in One Dimension

A Vibrating String

During this lecture and the next, we will investigate how to model the displacement of a point on a vibrating string (which will give us the one-dimensional wave equation), and then look at a technique that can be used to solve this PDE (along with many others).

Problem: Suppose we have an elastic string (eg a guitar string) supported between two fixed ends a distance L apart. Find a model to describe the displacement of the string, as a function of x (the distance along the string) and t (time).

A Vibrating String

The string is fixed at the ends, ie the points $x = 0$ and $x = L$. The ultimate aim is to find a function $u(x, t)$ that will describe the vertical displacement of a point on the string a distance x from the first fixed point, at time t .

Before we start looking at the model we will make some assumptions:

1. The string has the same physical properties (in particular mass per unit length) at every point along the string.
2. The string is perfectly elastic and doesn't offer any resistance to bending.
3. The tension in the string is large enough that the effect of gravity (pulling down the string) is negligible.
4. The points in the string move in small (compared to the length of the string) vertical motions.

A Vibrating String

We will now work through a derivation of the wave equation, which will be demonstrated in the lecture. The diagram given below may be found in your textbook (Kreyszig, p543).

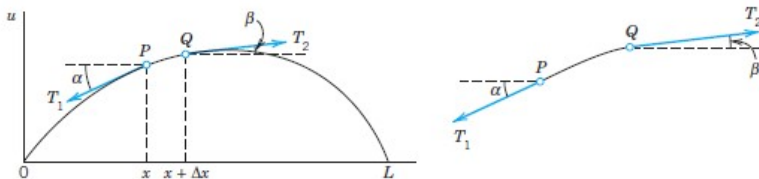


Fig. 286. Deflected string at fixed time t . Explanation on p. 544

A Vibrating String

We consider the forces acting on a small portion of the string, between x and $x + \Delta x$. Since the string offers no resistance to bending, the tension is tangential to the curve of the string at each point. Let T_1 and T_2 be the tension at the endpoints P and Q of that portion. Since the points of the string move vertically, there is no motion in the horizontal direction. Hence, the horizontal components of the tension must be constant. Hence,

$$T_1 \cos(\alpha) = T_2 \cos(\beta) = T.$$

In the vertical direction we have two forces, the downwards $-T_1 \sin(\alpha)$ and the upwards $T_2 \sin(\beta)$. The sum of these forces must be equal to the acceleration of the portion times the mass of the portion. If ρ is the mass per unit length of the string then

$$T_2 \sin(\beta) - T_1 \sin(\alpha) = \rho \Delta x \frac{\partial^2 u}{\partial t^2}$$

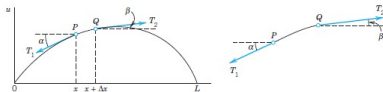


Fig. 286. Deflected string at fixed time t . Explanation on p. 544

A Vibrating String

Now divide each term in the equation

$$T_2 \sin(\beta) - T_1 \sin(\alpha) = \rho \Delta x \frac{\partial^2 u}{\partial t^2}$$

by T (and assume that angles β and α are small) to obtain the approximation

$$\frac{T_2 \sin(\beta)}{T_2 \cos(\beta)} - \frac{T_1 \sin(\alpha)}{T_1 \cos(\alpha)} = \frac{\rho \Delta x}{T} \frac{\partial^2 u}{\partial t^2}$$

and so

$$\tan(\beta) - \tan(\alpha) = \frac{\rho \Delta x}{T} \frac{\partial^2 u}{\partial t^2}$$

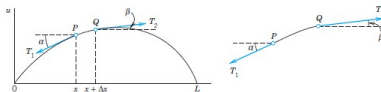


Fig. 286. Deflected string at fixed time t . Explanation on p. 544

A Vibrating String

Now

$$\tan(\beta) - \tan(\alpha) = \frac{\rho \Delta x}{T} \frac{\partial^2 u}{\partial t^2}$$

We have

$$\tan(\alpha) = \left(\frac{\partial u}{\partial x} \right) \Big|_x \quad \text{and} \quad \tan(\beta) = \left(\frac{\partial u}{\partial x} \right) \Big|_{x+\Delta x}$$

so

$$\left(\frac{\partial u}{\partial x} \right) \Big|_{x+\Delta x} - \left(\frac{\partial u}{\partial x} \right) \Big|_x = \frac{\rho \Delta x}{T} \frac{\partial^2 u}{\partial t^2}$$

Divide both sides by Δx and take the limit as $\Delta x \rightarrow 0$ to obtain

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

where $c^2 = \frac{T}{\rho}$ denotes a positive quantity. The equation is homogeneous and of second order.

Exercise

Work through this derivation yourself. Sketch the diagram.

Text Book References

Chapters 12.2, 12.3